

Tut Test 3

The Bode plot on the back of this page shows the frequency response of an unknown system.

- a) What is the order of the system? Explain your answer with reference to the plot. [2 marks]
- b) Using features of the plot, explain why it would be acceptable to model the system as second-order. For what range of frequencies would the second-order model hold? (You may give a loose range in terms of the nearest decade.)
[3 marks]
- c) Use the plot to estimate a second-order transfer function for the system. [5 marks]

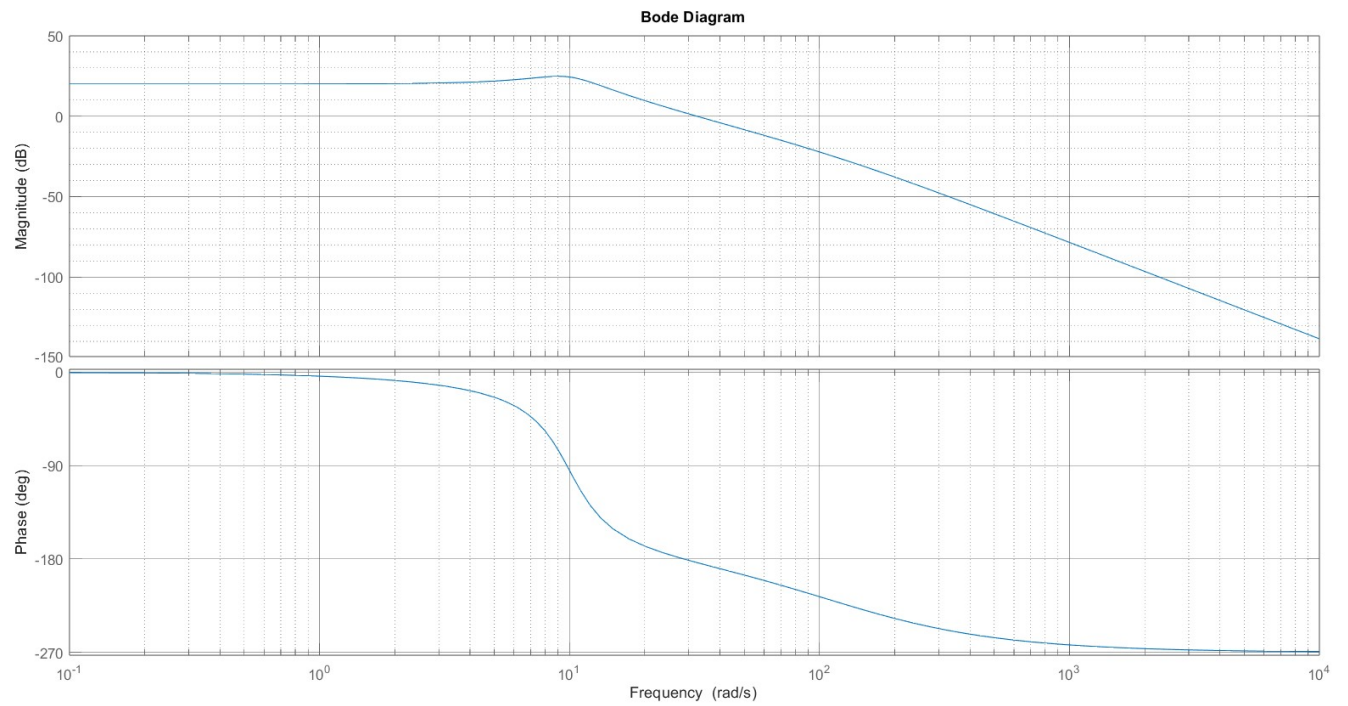


Figure 1: Bode plot for frequencies up to 10^4 rad/s.

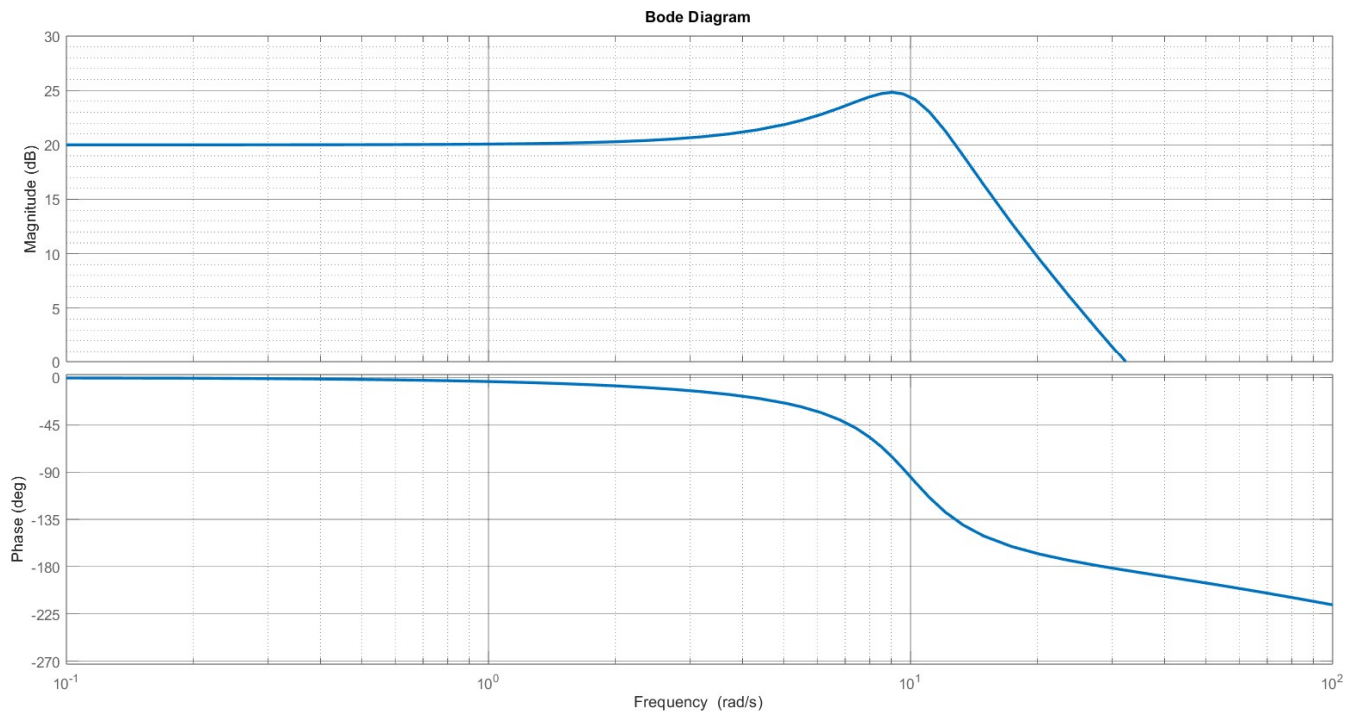


Figure 2: Zoomed in Bode plot for same system

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Answer sheet

Name: _____

Student No: _____

a)

b)

c)